

DATA ANALYTICS LAB

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5DS02	PCC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	30	70	100

COURSE OBJECTIVES

The course should enable the students to:

1. The objective of this course is to provide comprehensive knowledge of python programming paradigms required for Data Analytics.

COURSE OUTCOMES

At the end of the course, students will be able to:

1. Make efficient use of the data structures built into Python
2. Demonstrate handling of missing data
3. Implement data-analytic tasks in Python using external libraries such as Pandas
4. Implement numerical programming, data handling and visualization through NumPy, Pandas modules
5. Apply analytics for business situations

WEEK-I WORKING WITH R

- Reading different types of data sets (.txt, .csv) from web and disk and writing in file in specific disk location.
- Reading Excel data sheet in R.
- Reading XML dataset in R.

WEEK-2 LEARNING NUMPY

- Creating arrays
- Using arrays and scalars
- Indexing Arrays
- Transposition
- Array Processing
- Array Input and Output

WEEK-3 INTRODUCTION TO PANDAS PART-I

- Series
- DataFrames
- Index objects
- Reindex
- Drop Entry
- Selecting Entries

WEEK-4 INTRODUCTION TO PANDAS PART-II

- Data Alignment
- Rank and Sort
- Summary Statistics
- Missing Data
- Index Hierarchy

WEEK-5	WORKING WITH DATA PART-I
• Reading and Writing Text Files • JSON with Python • HTML with PythonMicrosoft • Excel files with Python • Microsoft Excel files with Python	
WEEK-6	WORKING WITH DATA PART-II
• Merge • Combining DataFrames • Pivoting • Duplicates in DataFrames • Mapping • Binning • Outliers	
WEEK-7	WORKING WITH DATA PART-III
• GroupBy on DataFrames • Splitting Applying and Combining • Cross Tabulation	
WEEK-8	DATA VISUALIZATION
• Installing Seaborn • Histograms • Combining Plot Styles	
WEEK-9	DATA VISUALIZATION
• Box and Violin Plots • Regression Plots • Heatmaps and Clustered Matrice • TabPy: Tableau with python	
WEEK-10	USE CASES
• Data Project - Stock Market Analysis • Data Project - Customer Segmentation • Data Project - Credit Card Fraud Detection	
TEXT BOOKS	
1. DJANGO Web Development with python by by Arun Ravindran Samuel Dauzon Aidas Bendoraitls, 2017 2. Django for Beginners : Build Websites with Python and Django by by William S Vincent 2020	
REFERENCE BOOKS	
1. Lightweight Django by Julia Elman,2014 2. Django Design patterns and Best Practices – second Edition by Arun Ravindran ,2018	